



Image



Title
LF353

FigureCaption
SLOS012C – MARCH 1987 – REVISED MARCH 2016

LF353 Wide-Bandwidth JFET-Input Dual Operational Amplifier

Title

1 Features

ListItem

- Low Input Bias Current 50 pA Typical
- Low Input Noise Current 0.01 pA/√Hz Typical
- Low Supply Current 3.6 mA Typical
- High Input Impedance $10^{12} \Omega$ Typical
- Internally-Trimmed Offset Voltage
- Gain Bandwidth 3 MHz Typical
- High Slew Rate 13 V/μs Typical

Title

2 Applications

ListItem

- Motor Integrated Systems: UPS
- Drives and Control Solutions: AC Inverter and VF Drives
- Renewables: Solar Inverters
- Pro Audio Mixers
- Oscilloscopes

Title

3 Description

NarrativeText

This LF353 device is a low-cost, high-speed, JFET-input operational amplifier with very low input offset voltage. It requires low supply current yet maintains a large gain-bandwidth product and a fast slew rate. In addition, the matched high-voltage JFET input provides very low input bias and offset currents.

NarrativeText

The LF353 can be used in applications such as high-speed integrators, digital-to-analog converters, sample-and-hold circuits, and many other circuits.

NarrativeText

The LF353 is characterized for operation from 0°C to 70°C.

FigureCaption

Device Information⁽¹⁾

Table

PART NUMBER	PACKAGE	BODY SIZE (NOM)
LF353D	SOIC (8)	4.90 mm × 3.91 mm
LF353P	PDIP (8)	9.81 mm × 6.35 mm

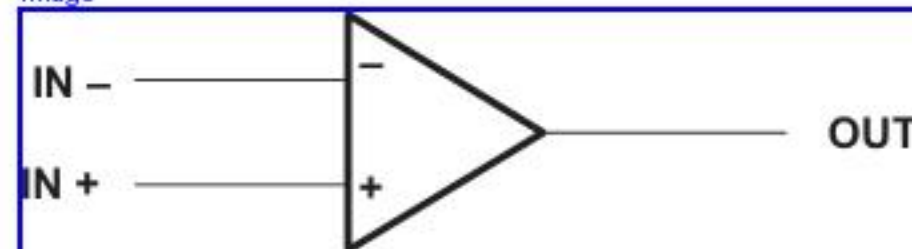
ListItem

(1) For all available packages, see the orderable addendum at the end of the data sheet.

UncategorizedText

Symbol

Image



NarrativeText



An IMPORTANT NOTICE at the end of this data sheet addresses availability, warranty, changes, use in safety-critical applications, intellectual property matters and other important disclaimers. PRODUCTION DATA.

Title

Table of Contents

Table

1 Features	1
2 Applications	1
3 Description	1
4 Revision History	2
5 Pin Configuration and Functions	3
6 Specifications	4
6.1 Absolute Maximum Ratings	4
6.2 ESD Ratings	4
6.3 Recommended Operating Conditions	4
6.4 Thermal Information	4
6.5 Electrical Characteristics	5
6.6 Switching Characteristics	5
6.7 Typical Characteristics	6
7 Parameter Measurement Information	7
8 Detailed Description	8
8.1 Overview	8
8.2 Functional Block Diagram	8

Table

8.3 Feature Description	8
8.4 Device Functional Modes	8
9 Application and Implementation	9
9.1 Application Information	9
9.2 Typical Application	9
10 Power Supply Recommendations	10
11 Layout	11
11.1 Layout Guidelines	11
11.2 Layout Example	11
12 Device and Documentation Support	12
12.1 Documentation Support	12
12.2 Community Resources	12
12.3 Trademarks	12
12.4 Electrostatic Discharge Caution	12
12.5 Glossary	12
13 Mechanical, Packaging, and Orderable Information	12

Title

4 Revision History

Narrative Text

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Title

Changes from Revision B (August 1994) to Revision C

UncategorizedText

Page

List Item

Added *ESD Ratings* table, *Feature Description* section, *Device Functional Modes*, *Application and Implementation* section, *Power Supply Recommendations* section, *Layout* section, *Device and Documentation Support* section, and *Mechanical, Packaging, and Orderable Information* section.

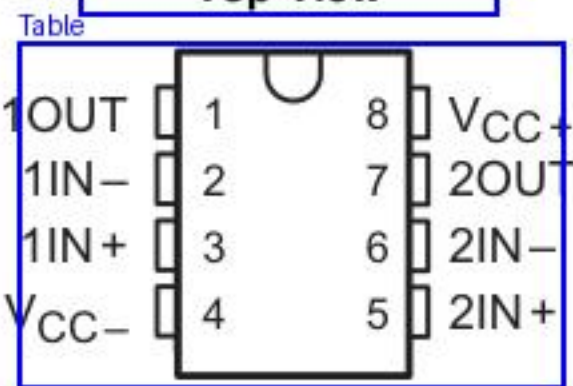
UncategorizedText

Title

5 Pin Configuration and Functions

FigureCaption

D or P Package
8-Pin SOIC or PDIP
Top View



FigureCaption

Pin Functions

Table

PIN		I/O	DESCRIPTION
NAME	NO.		
1OUT	1	O	Output
1IN-	2	I	Inverting input
1IN+	3	I	Noninverting input
V _{CC} -	4	—	Negative supply voltage
2IN+	5	I	Noninverting input
2IN-	6	I	Inverting input
2OUT	7	O	Output
V _{CC} +	8	—	Positive supply voltage

Title

6 Specifications

Title

6.1 Absolute Maximum Ratings

Narrative Text

over operating free-air temperature range (unless otherwise noted)⁽¹⁾

		MIN	MAX	UNIT
V _{CC+}	Supply voltage		18	V
V _{CC-}	Supply voltage		–18	V
VID	Differential input voltage		±30	V
VI	Input voltage ⁽²⁾		±15	V
	Duration of output short circuit		Unlimited	s
	Continuous total power dissipation		500	mW
	Lead temperature 1.6 mm (1/16 inch) from case for 10 s		260	°C
T _J	Junction temperature		150	°C
T _{stg}	Storage temperature	–65	150	°C

List Item

- (1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
- (2) Unless otherwise specified, the absolute maximum negative input voltage is equal to the negative power supply voltage.

Title

6.2 ESD Ratings

		VALUE	UNIT
V _(ESD)	Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001 ⁽¹⁾	±2000
		Charged-device model (CDM), per JEDEC specification JESD22-C101 ⁽²⁾	±1000

List Item

- (1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.
- (2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

Title

6.3 Recommended Operating Conditions

Narrative Text

over operating free-air temperature range (unless otherwise noted)

		MIN	MAX	UNIT
V _{CC+}	Supply voltage	3.5	18	V
V _{CC-}	Supply voltage	–3.5	–18	V
V _{CM}	Common-mode voltage	V _{CC-} + 4	V _{CC+} – 4	V
T _A	Operating temperature	0	70	°C

Title

6.4 Thermal Information

THERMAL METRIC ⁽¹⁾		LF353		UNIT
		D (SOIC)	P (PDIP)	
		8 PINS	8 PINS	
R _{θJA}	Junction-to-ambient thermal resistance	106.6	55.1	°C/W
R _{θJC(top)}	Junction-to-case (top) thermal resistance	51.5	45	°C/W
R _{θJB}	Junction-to-board thermal resistance	46.5	32.2	°C/W
ψ _{JT}	Junction-to-top characterization parameter	9.8	22.6	°C/W
ψ _{JB}	Junction-to-board characterization parameter	46.1	32.2	°C/W

Narrative Text

- (1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application report, [SPRA953](#).

List Item

4 [Submit Documentation Feedback](#)

Narrative Text

Product Folder Links: [LF353](#)

Narrative Text

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6.5 Electrical Characteristics

FigureCaption

Table

$T_A = 0^\circ\text{C to } 70^\circ\text{C}$, $V_{CC+} = \pm 15\text{ V}$ (unless otherwise noted)

PARAMETER		TEST CONDITIONS		MIN	TYP	MAX	UNIT
V_{IO}	Input offset voltage	$V_{IC} = 0$, $R_S = 10\text{ k}\Omega$	$T_A = 25^\circ\text{C}$		5	10	mV
			Full range ⁽¹⁾			13	
α_{VIO}	Average temperature coefficient of inputs offset voltage	$V_{IC} = 0$, $R_S = 10\text{ k}\Omega$			10		$\mu\text{V}/^\circ\text{C}$
I_{IO}	Input offset current ⁽²⁾	$V_{IC} = 0$	$T_A = 25^\circ\text{C}$		25	100	pA
			$T_A = 70^\circ\text{C}$			4	nA
I_{IB}	Input bias current ⁽²⁾	$V_{IC} = 0$	$T_A = 25^\circ\text{C}$		50	200	pA
			$T_A = 70^\circ\text{C}$			8	nA
V_{ICR}	Common-mode input voltage range	Lower limit of range		-11	-12		V
		Upper limit of range		11	15		
V_{OM}	Maximum peak output voltage swing	$R_L = 10\text{ k}\Omega$		± 12	± 13.5		V
A_{VD}	Large-signal differential voltage	$V_O = \pm 10\text{ V}$, $R_L = 2\text{ k}\Omega$	$T_A = 25^\circ\text{C}$	25	100		V/mV
			Full range ⁽¹⁾	15			
r_i	Input resistance	$T_J = 25^\circ\text{C}$			10^{12}		Ω
CMRR	Common-mode rejection ratio	$R_S \leq 10\text{ k}\Omega$		70	100		dB
k_{SVR}	Supply-voltage rejection ratio	See ⁽³⁾		70	100		dB
I_{CC}	Supply current				3.6	6.5	mA

Narrative Text

(1) Full range is $0^\circ\text{C to } 70^\circ\text{C}$

(2) Input bias currents of a FET-input operational amplifier are normal junction reverse currents, which are temperature sensitive. Pulse

techniques must be used that will maintain the junction temperatures as close to the ambient temperature as possible

(3) Supply-voltage rejection ratio is measured for both supply magnitudes increasing or decreasing simultaneously.

6.6 Switching Characteristics

FigureCaption

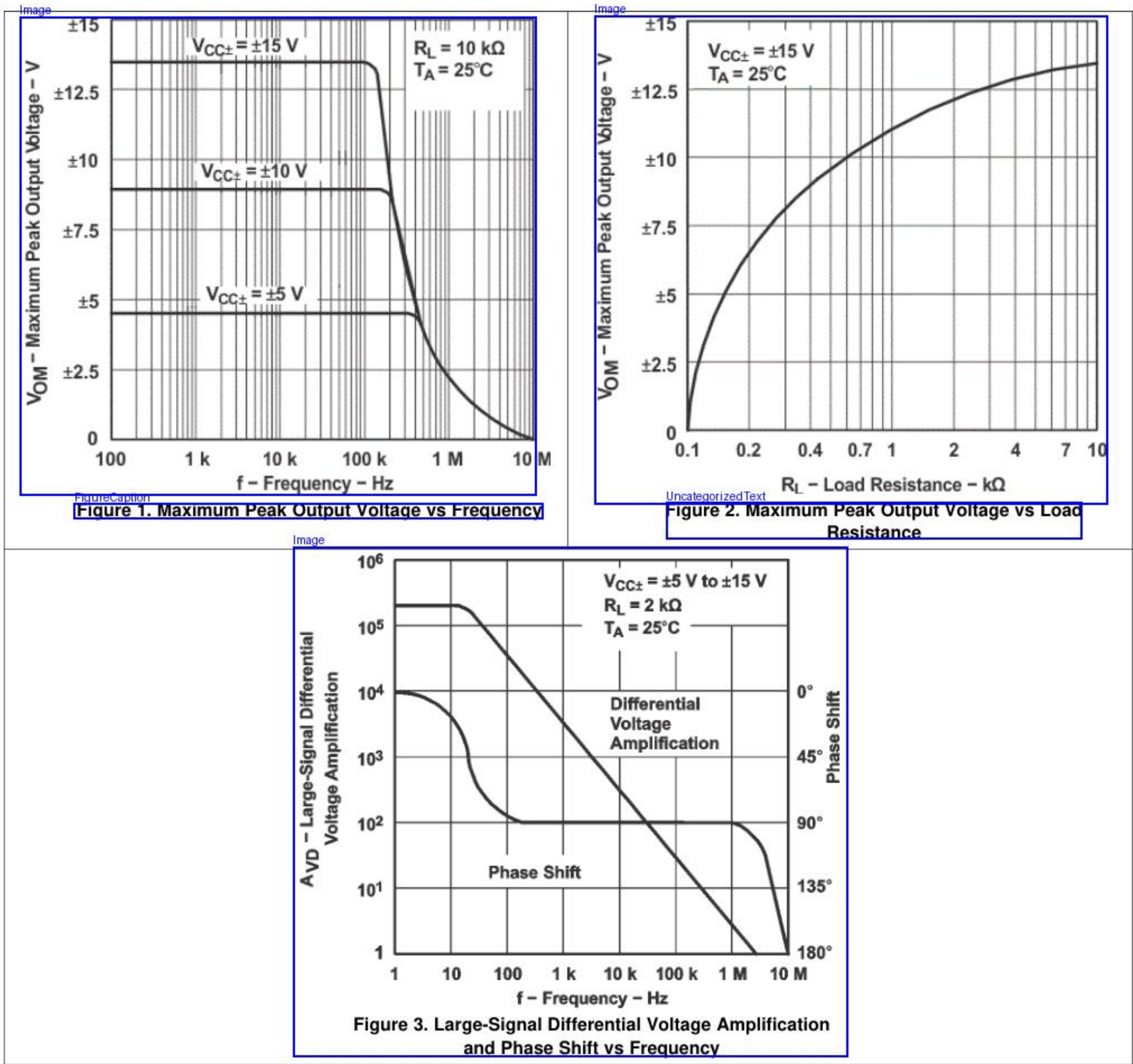
Table

$V_{CC+} = \pm 15\text{ V}$, $T_A = 25^\circ\text{C}$, over operating free-air temperature range (unless otherwise noted)

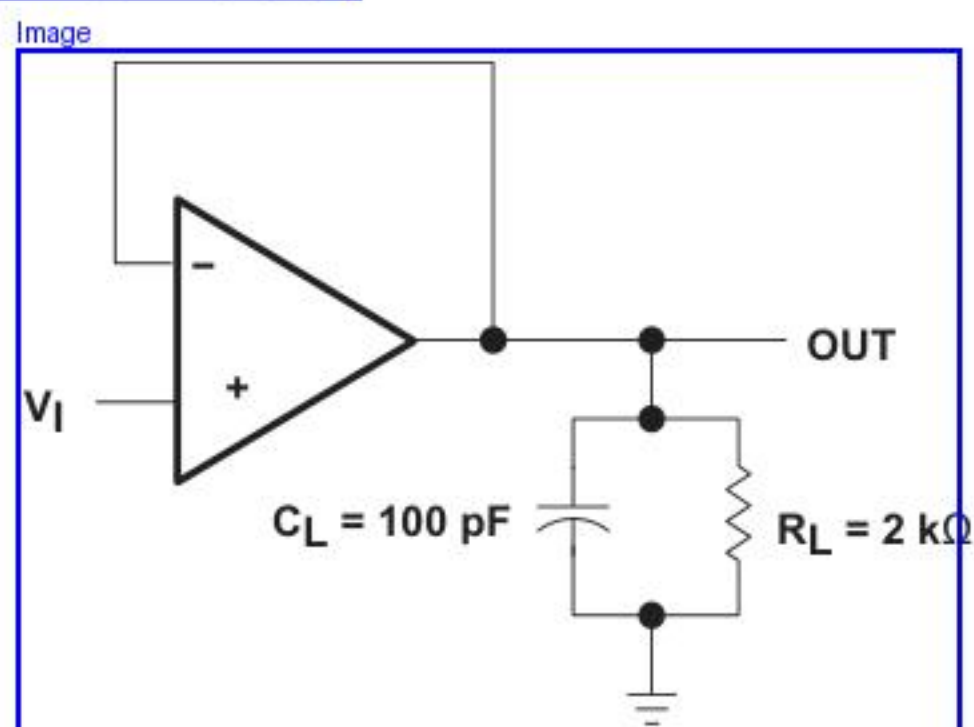
PARAMETER		TEST CONDITIONS		MIN	TYP	MAX	UNIT
V_{O1}/V_{O2}	Crosstalk attenuation	$f = 1\text{ kHz}$			120		dB
SR	Slew rate			8	13		V/ μs
B1	Unity-gain bandwidth				3		MHz
V_n	Equivalent input noise voltage	$f = 1\text{ kHz}$, $R_S = 20\text{ }\Omega$			18		nV/ $\sqrt{\text{Hz}}$
I_n	Equivalent input noise current	$f = 1\text{ kHz}$			0.01		pA/ $\sqrt{\text{Hz}}$

Title

6.7 Typical Characteristics



Title
7 Parameter Measurement Information



FigureCaption
Figure 4. Unity-Gain Amplifier

Title

LF353

SLOS012C –MARCH 1987–REVISED MARCH 2016

Title

8 Detailed Description

Title

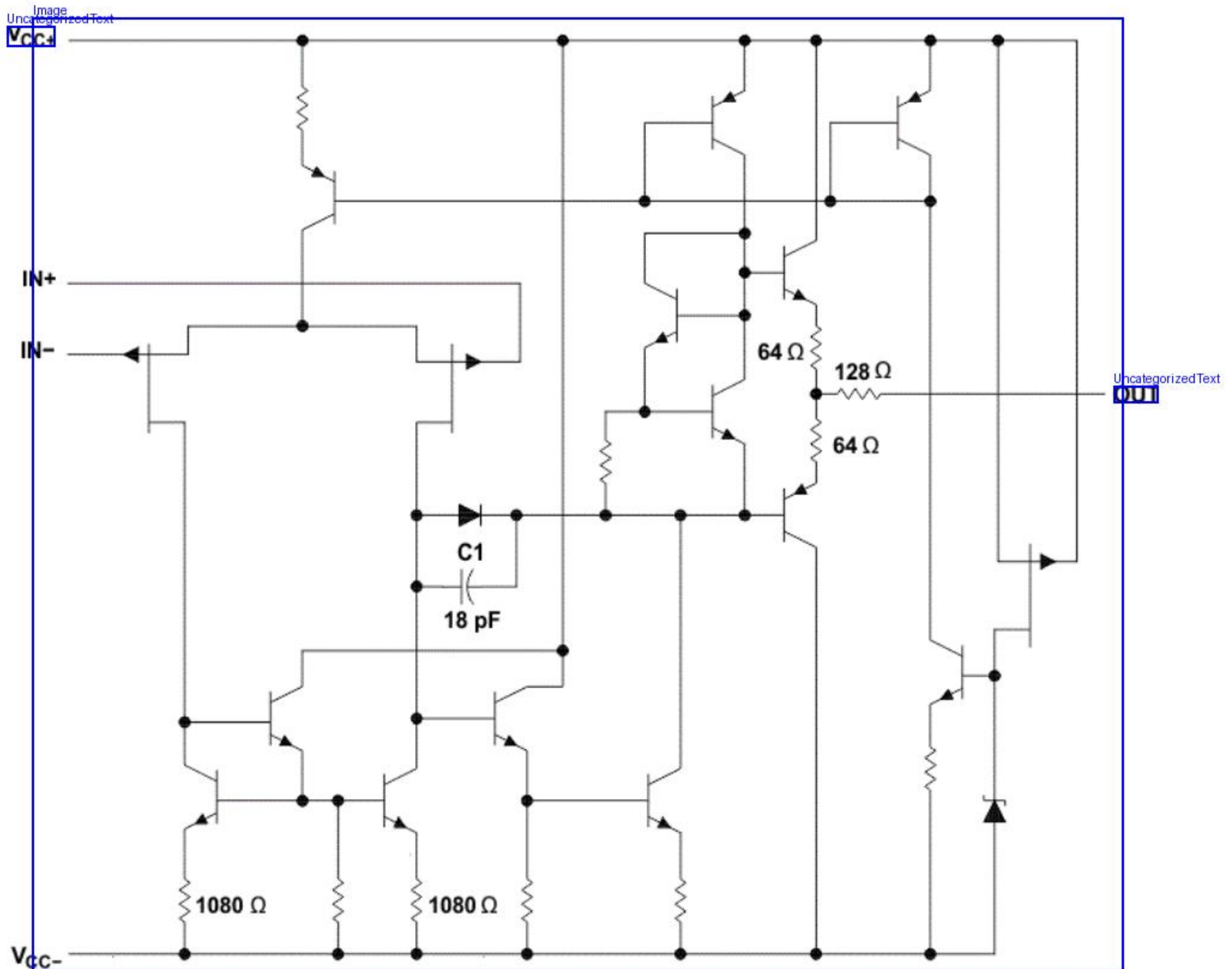
8.1 Overview

NarrativeText

The LF353 device is a JFET-input operational amplifier with low input bias and offset currents and fast slew rate. Each amplifier features JFET inputs (for high input impedance) coupled with bipolar output stages integrated on a single monolithic chip. The output is protected against shorts due to the resistive 200-Ω output impedance.

Title

8.2 Functional Block Diagram



Title

8.3 Feature Description

Title

8.3.1 Slew Rate

NarrativeText

The slew rate is the rate at which an operational amplifier can change its output when there is a change on the input. These devices have a 13-V/μs slew rate.

Title

8.4 Device Functional Modes

NarrativeText

These devices are powered on when the supply is connected. This device can be operated as a single-supply operational amplifier or dual-supply amplifier depending on the application.

Footer

8 Submit Documentation Feedback

NarrativeText

Product Folder Links: LF353

NarrativeText

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Title 9 Application and Implementation

Title NOTE

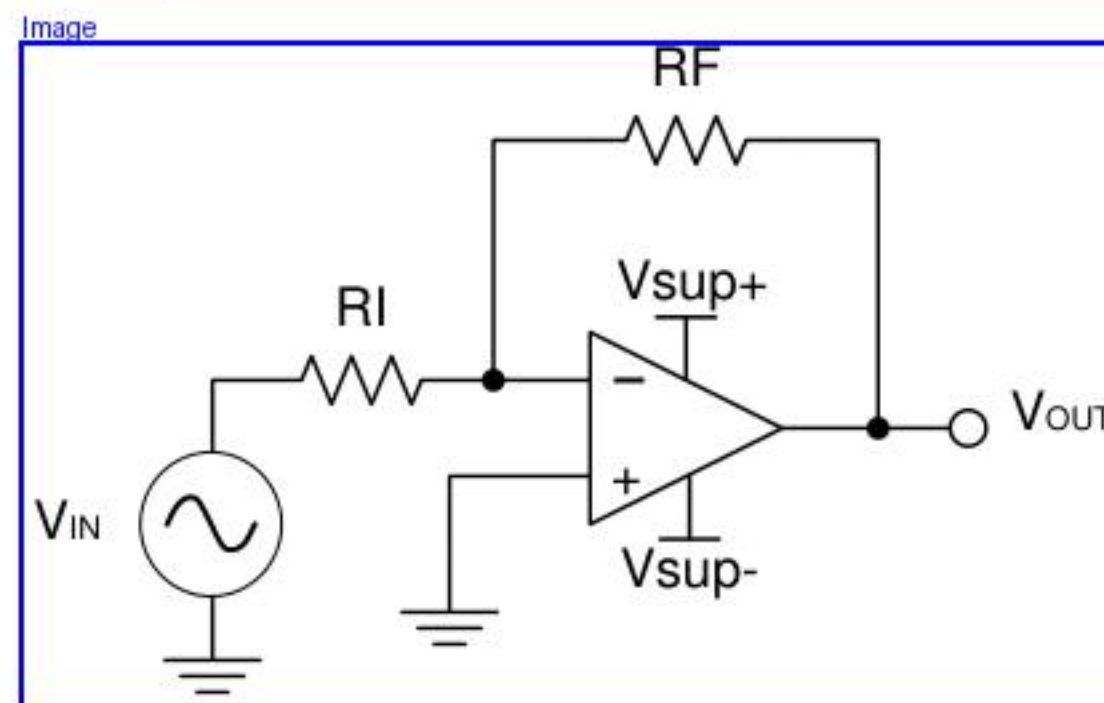
NarrativeText
Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

Title 9.1 Application Information

NarrativeText
The LF353 has two independent amplifiers that have very low input bias current which allow using higher resistance resistors in the feedback network. The upper input common mode range goes to the upper supply rail. The lower common mode range does not include the negative supply rail. Output resistance is 200 ohms to protect the device from accidental shorts.

Title 9.2 Typical Application

NarrativeText
A typical application for an operational amplifier is an inverting amplifier. This amplifier takes a positive voltage on the input, and makes it a negative voltage. In the same manner, it also makes negative voltages positive.



FigureCaption
Figure 5. Inverting Amplifier

Title 9.2.1 Design Requirements

NarrativeText
The supply voltage must be chosen such that it is larger than the input voltage range and output range. For instance, this application scales a signal of ± 0.5 V to ± 1.8 V. Setting the supply at ± 12 V is sufficient to accommodate this application.

Title 9.2.2 Detailed Design Procedure

NarrativeText
Determine the gain required by the inverting amplifier using [Equation 1](#) and [Equation 2](#).

Formula

$$A_v = \frac{V_{OUT}}{V_{IN}}$$

$$A_v = \frac{1.8}{-0.5} = -3.6$$

UncategorizedText
1

NarrativeText
(2)

NarrativeText
Once the desired gain is determined, choose a value for R_I or R_F . Choosing a value in the k Ω range is desirable because the amplifier circuit uses currents in the mA range. This ensures the part does draw too much current. For this example, choose 10 k Ω for R_I and 36 k Ω for R_F , as shown in [Equation 3](#).

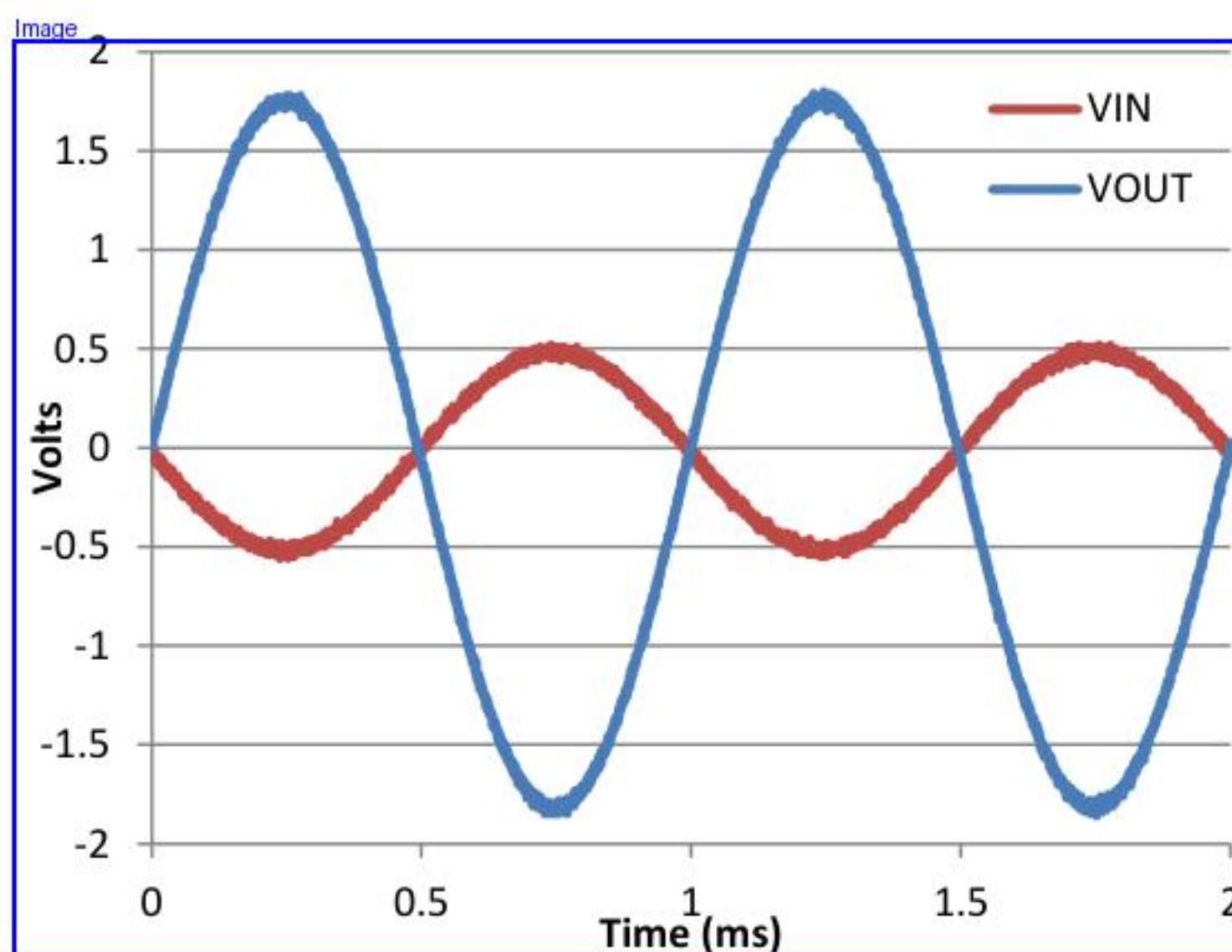
Formula

$$A_v = -\frac{R_F}{R_I}$$

UncategorizedText
3

Typical Application (continued)

9.2.3 Application Curve



FigureCaption

Figure 6. Input and Output Voltages of the Inverting Amplifier

10 Power Supply Recommendations

CAUTION

Supply voltages larger than 36 V for a single-supply or outside the range of ± 18 V for a dual-supply can permanently damage the device (see the [Absolute Maximum Ratings](#)).

Place 0.1- μ F bypass capacitors close to the power-supply pins to reduce errors coupling in from noisy or high-impedance power supplies. For more detailed information on bypass capacitor placement, see the [Layout Example](#).

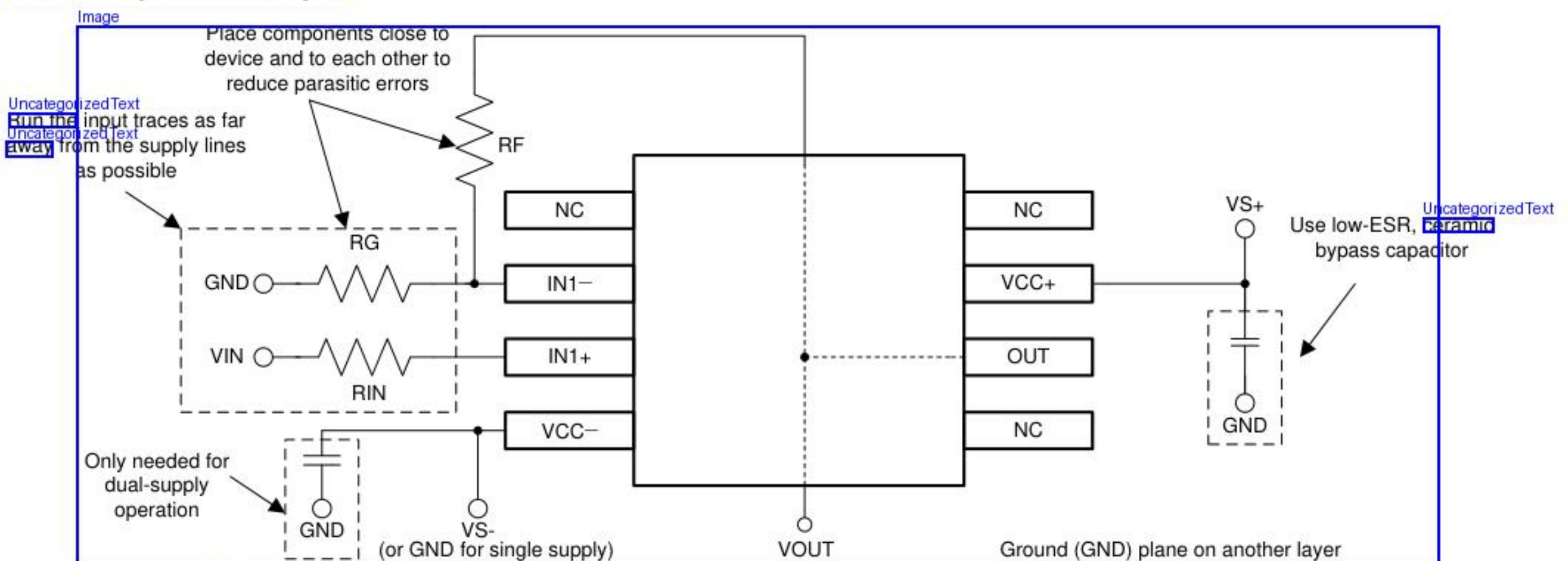
11 Layout

11.1 Layout Guidelines

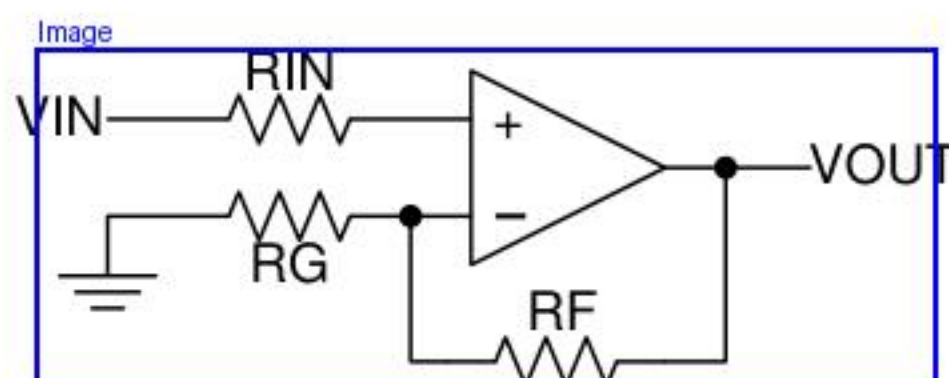
For best operational performance of the device, use the following layout guidelines:

- Noise can propagate into analog circuitry through the power pins of the circuit as a whole, as well as the operational amplifier. Bypass capacitors are used to reduce the coupled noise by providing low impedance power sources local to the analog circuitry.
 - Connect low-ESR, 0.1- μ F ceramic bypass capacitors between each supply pin and ground, placed as close to the device as possible. A single bypass capacitor from V+ to ground is applicable for single supply applications.
- Separate grounding for analog and digital portions of circuitry is one of the simplest and most-effective methods of noise suppression. One or more layers on multilayer PCBs are usually devoted to ground planes. A ground plane helps distribute heat and reduces EMI noise pickup. Make sure to physically separate digital and analog grounds, paying attention to the flow of the ground current. For more detailed information, see *Circuit Board Layout Techniques* (SLOA089).
- To reduce parasitic coupling, run the input traces as far away from the supply or output traces as possible. If it is not possible to keep them separate, it is much better to cross the sensitive trace perpendicular as opposed to in parallel with the noisy trace.
- Place the external components as close to the device as possible. Keeping RF and RG close to the inverting input minimizes parasitic capacitance, as shown in *Layout Example*.
- Keep the length of input traces as short as possible. Always remember that the input traces are the most sensitive part of the circuit.
- Consider a driven, low-impedance guard ring around the critical traces. A guard ring can significantly reduce leakage currents from nearby traces that are at different potentials.

11.2 Layout Example



FigureCaption
Figure 7. Operational Amplifier Board Layout for Noninverting Configuration



FigureCaption
Figure 8. Operational Amplifier Schematic for Noninverting Configuration

Title

12 Device and Documentation Support

Title

12.1 Documentation Support

Title

12.1.1 Related Documentation

Narrative Text

For related documentation see *Circuit Board Layout Techniques* (SLOA089).

Title

12.2 Community Resources

Narrative Text

The following links connect to TI community resources. Linked contents are provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's [Terms of Use](#).

UncategorizedText

E2E™ Online Community *TI's Engineer-to-Engineer (E2E) Community*. Created to foster collaboration among engineers. At e2e.ti.com, you can ask questions, share knowledge, explore ideas and help solve problems with fellow engineers.

Narrative Text

Design Support *TI's Design Support* Quickly find helpful E2E forums along with design support tools and contact information for technical support.

Title

12.3 Trademarks

Narrative Text

E2E is a trademark of Texas Instruments.
 All other trademarks are the property of their respective owners.

Title

12.4 Electrostatic Discharge Caution

UncategorizedText

Narrative Text

 These devices have limited built-in ESD protection. The leads should be shorted together or the device placed in conductive foam during storage or handling to prevent electrostatic damage to the MOS gates.

Title

12.5 Glossary

Narrative Text

SLYZ022 — *TI Glossary*.

Narrative Text

This glossary lists and explains terms, acronyms, and definitions.

Title

13 Mechanical, Packaging, and Orderable Information

Narrative Text

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

Title

PACKAGING INFORMATION

Table

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
LF353DR	Active	Production	SOIC (D) 8	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	LF353
LF353DR.A	Active	Production	SOIC (D) 8	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	0 to 70	LF353
LF353P	Active	Production	PDIP (P) 8	50 TUBE	Yes	NIPDAU	N/A for Pkg Type	0 to 70	LF353P
LF353P.A	Active	Production	PDIP (P) 8	50 TUBE	Yes	NIPDAU	N/A for Pkg Type	0 to 70	LF353P
LF353PE4	Active	Production	PDIP (P) 8	50 TUBE	-	Call TI	Call TI	0 to 70	

Title

(1) **Status:** For more details on status, see our [product life cycle](#).

Narrative Text

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

Narrative Text

(3) **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

Narrative Text

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

Narrative Text

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

Narrative Text

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Narrative Text

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

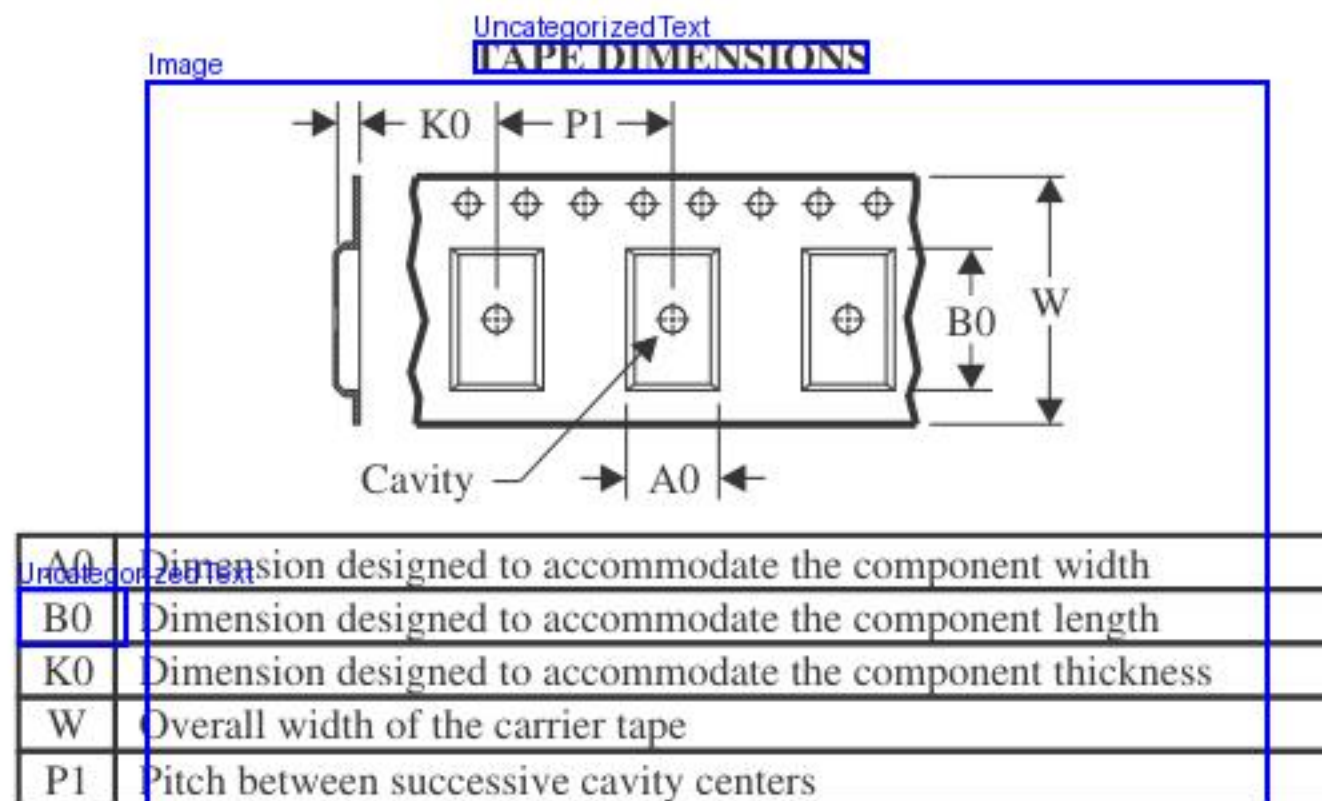
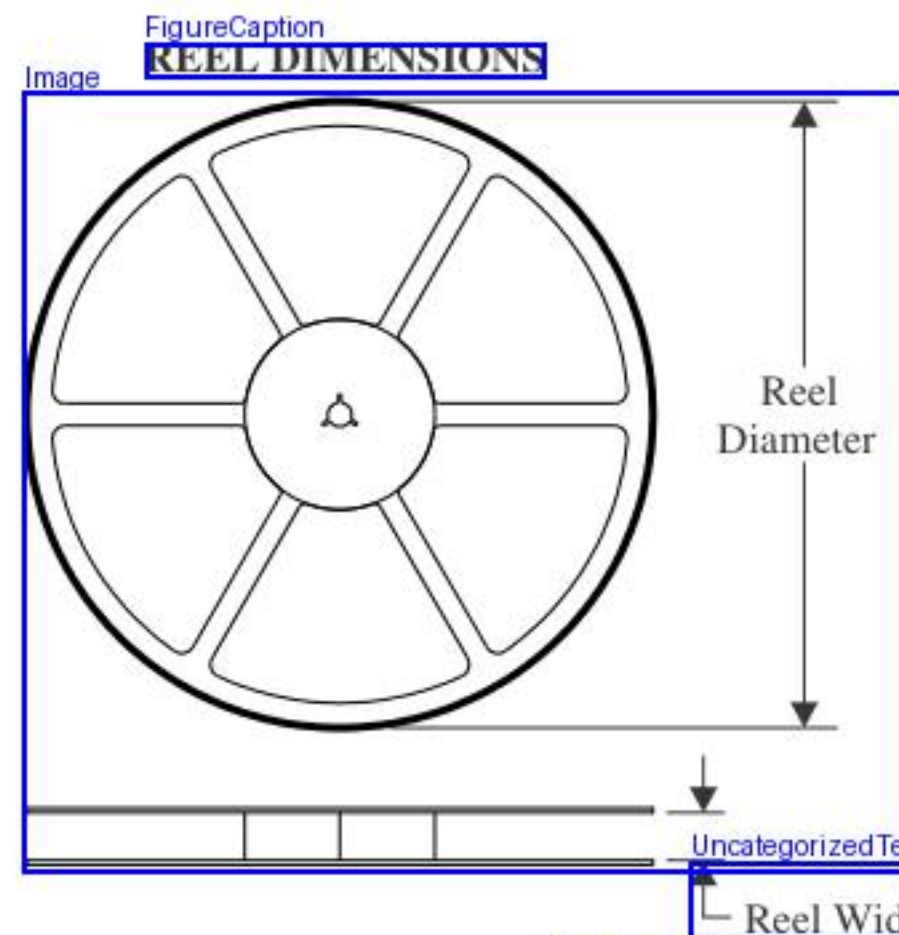
Narrative Text

Important Information and Disclaimer:The information provided on this page represents TI's knowledge and belief as of the date that it is provided. TI bases its knowledge and belief on information provided by third parties, and makes no representation or warranty as to the accuracy of such information. Efforts are underway to better integrate information from third parties. TI has taken and continues to take reasonable steps to provide representative and accurate information but may not have conducted destructive testing or chemical analysis on incoming materials and chemicals. TI and TI suppliers consider certain information to be proprietary, and thus CAS numbers and other limited information may not be available for release.

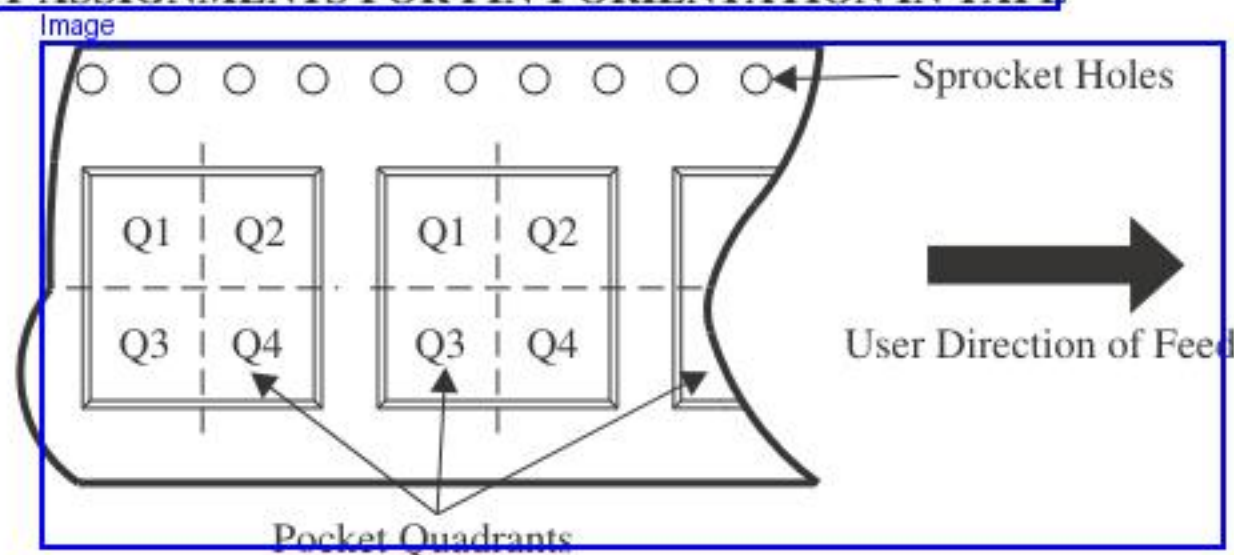
Narrative Text

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Title
TAPE AND REEL INFORMATION

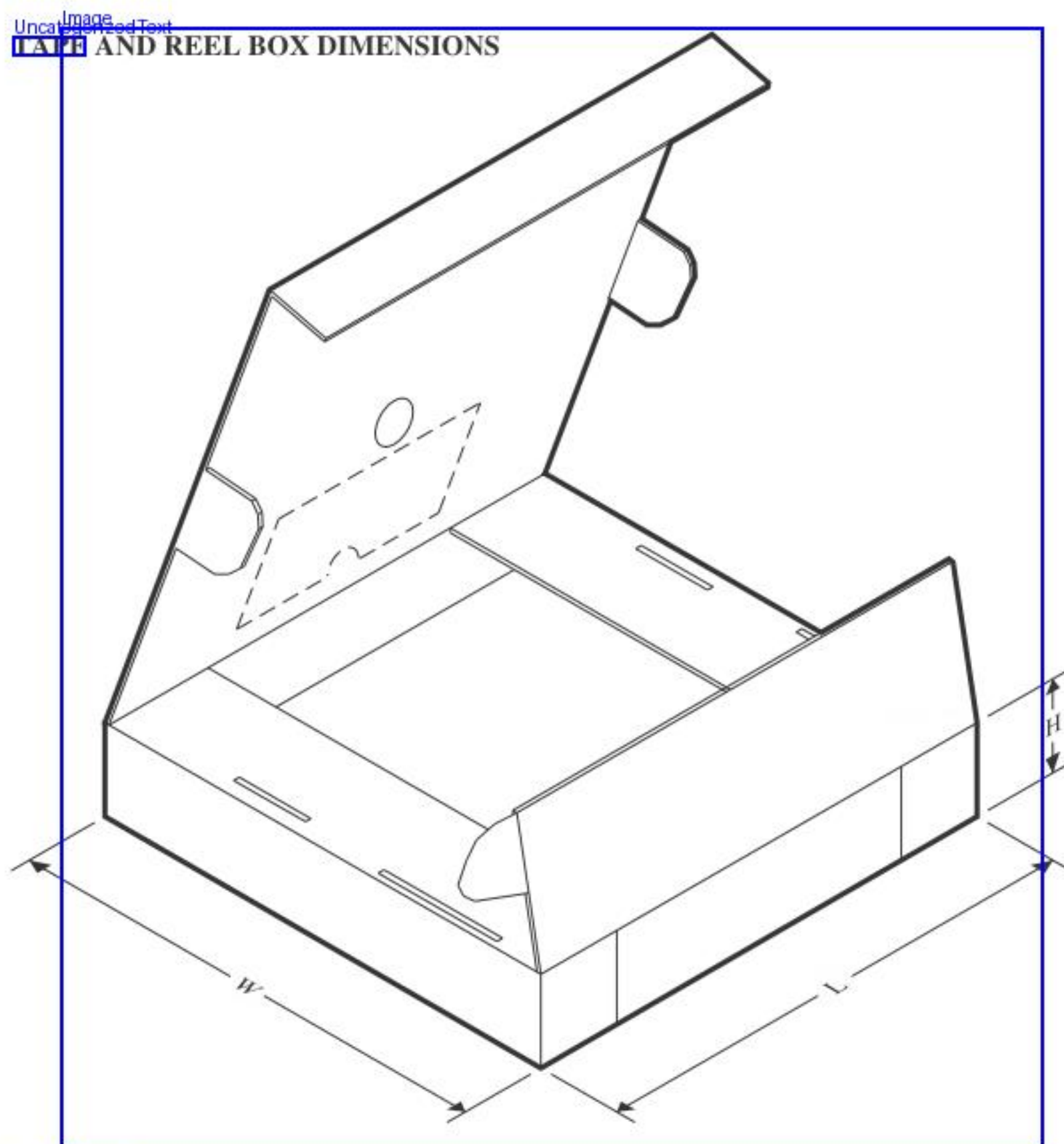


NarrativeText
QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



FigureCaption
 Table All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
LF353DR	SOIC	D	8	2500	330.0	12.4	6.4	5.2	2.1	8.0	12.0	Q1
LF353DR	SOIC	D	8	2500	330.0	12.4	6.5	5.4	2.0	8.0	12.0	Q1



UncategorizedText

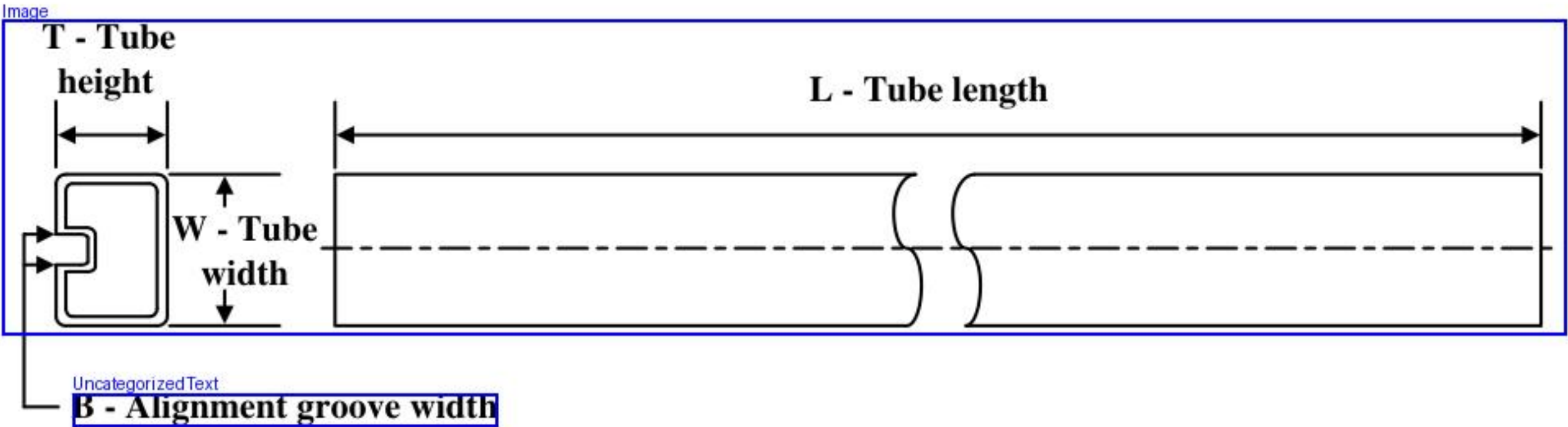
Table

All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
LF353DR	SOIC	D	8	2500	353.0	353.0	32.0
LF353DR	SOIC	D	8	2500	367.0	367.0	35.0

Title

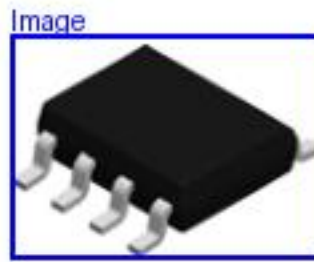
TUBE



UncategorizedText

TableAll dimensions are nominal

Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (μm)	B (mm)
LF353P	P	PDIP	8	50	506	13.97	11230	4.32
LF353P.A	P	PDIP	8	50	506	13.97	11230	4.32



Title

PACKAGE OUTLINE

Title

D0008A

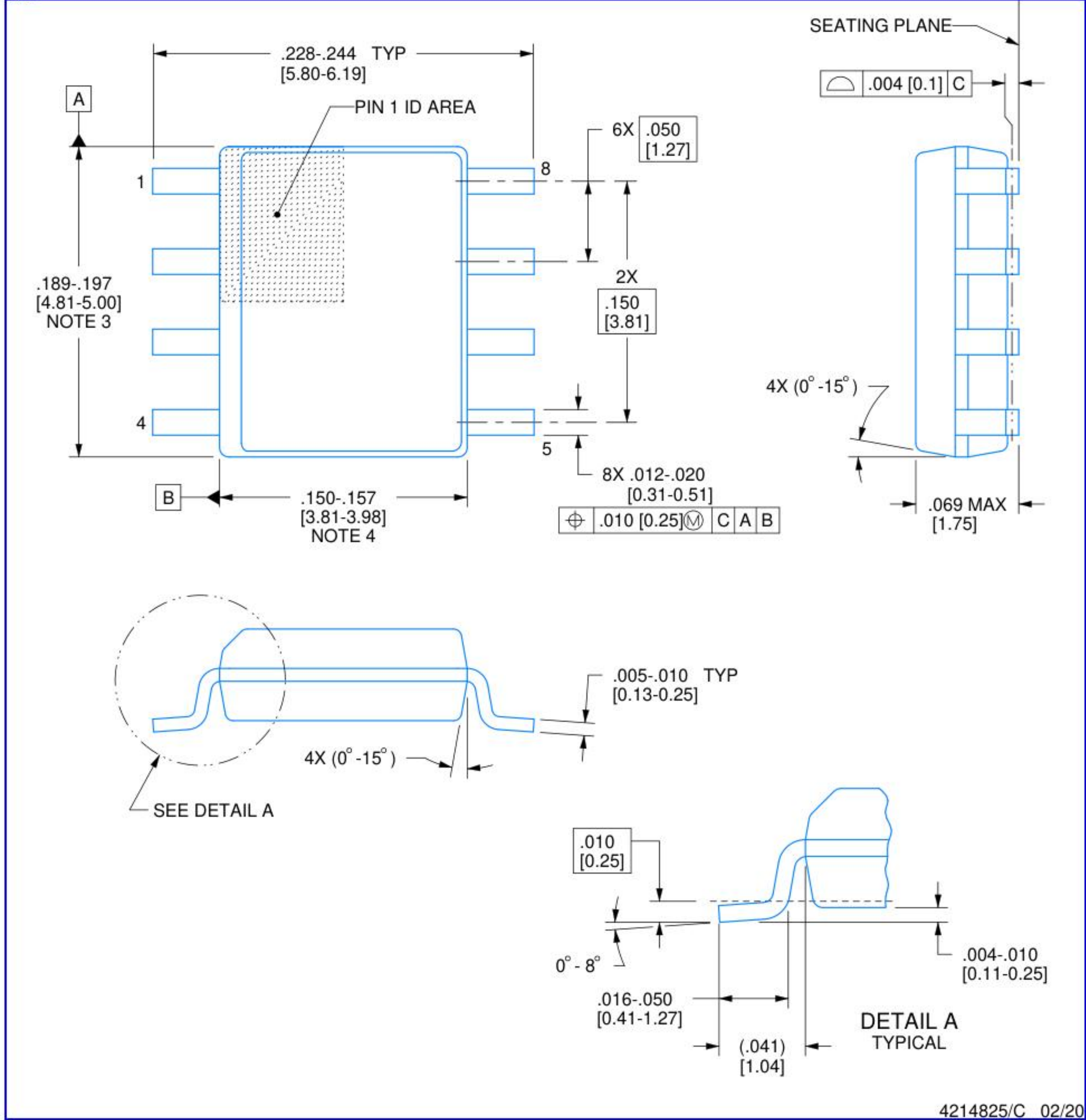
Title

SOIC - 1.75 mm max height

Narrative Text

SMALL OUTLINE INTEGRATED CIRCUIT

Image



4214825/C 02/2019

Narrative Text

NOTES:

List Item

1. Linear dimensions are in inches [millimeters]. Dimensions in parenthesis are for reference only. Controlling dimensions are in inches. Dimensioning and tolerancing per ASME Y14.5M.

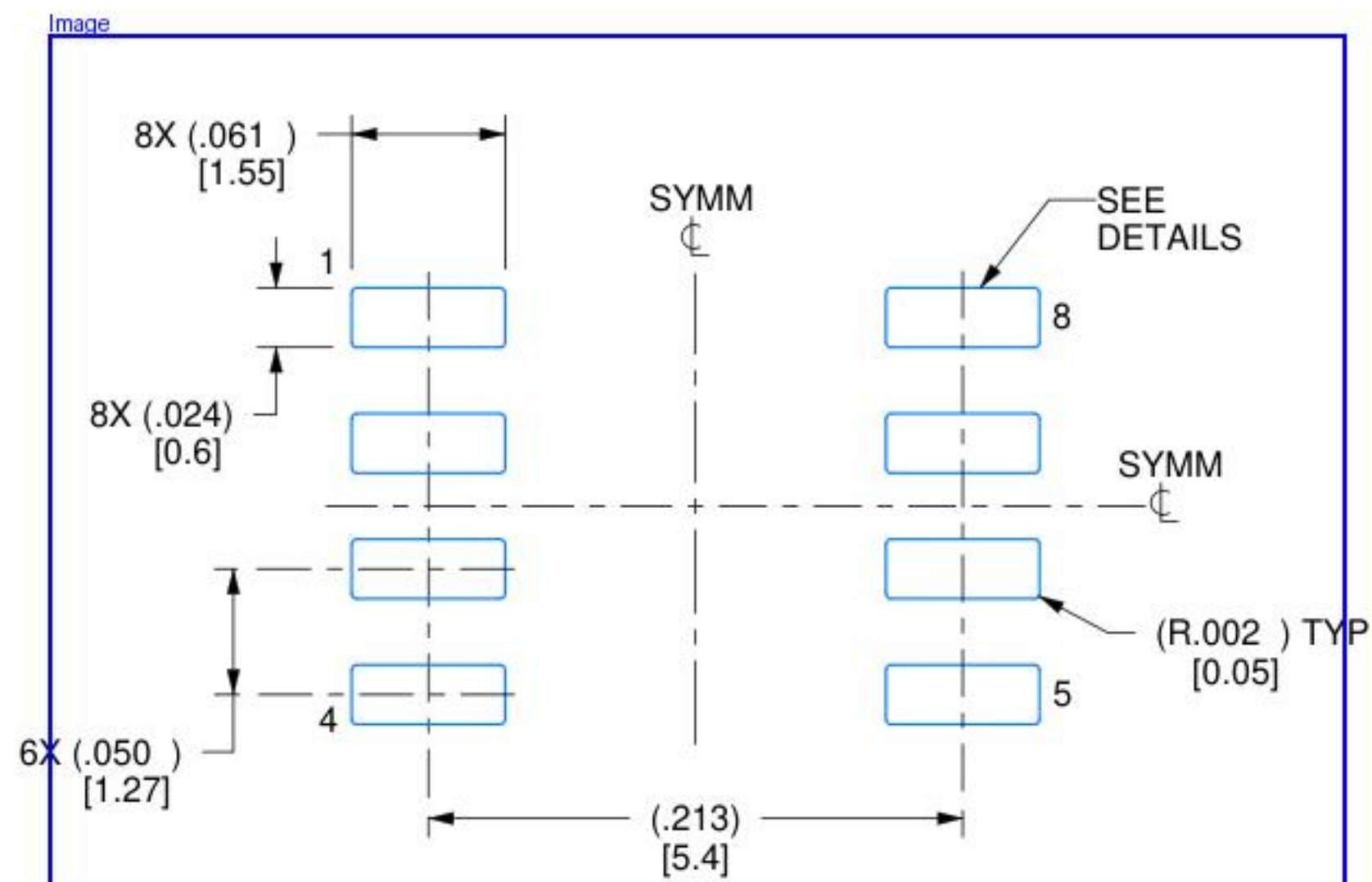
2. This drawing is subject to change without notice.

3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed .006 [0.15] per side.

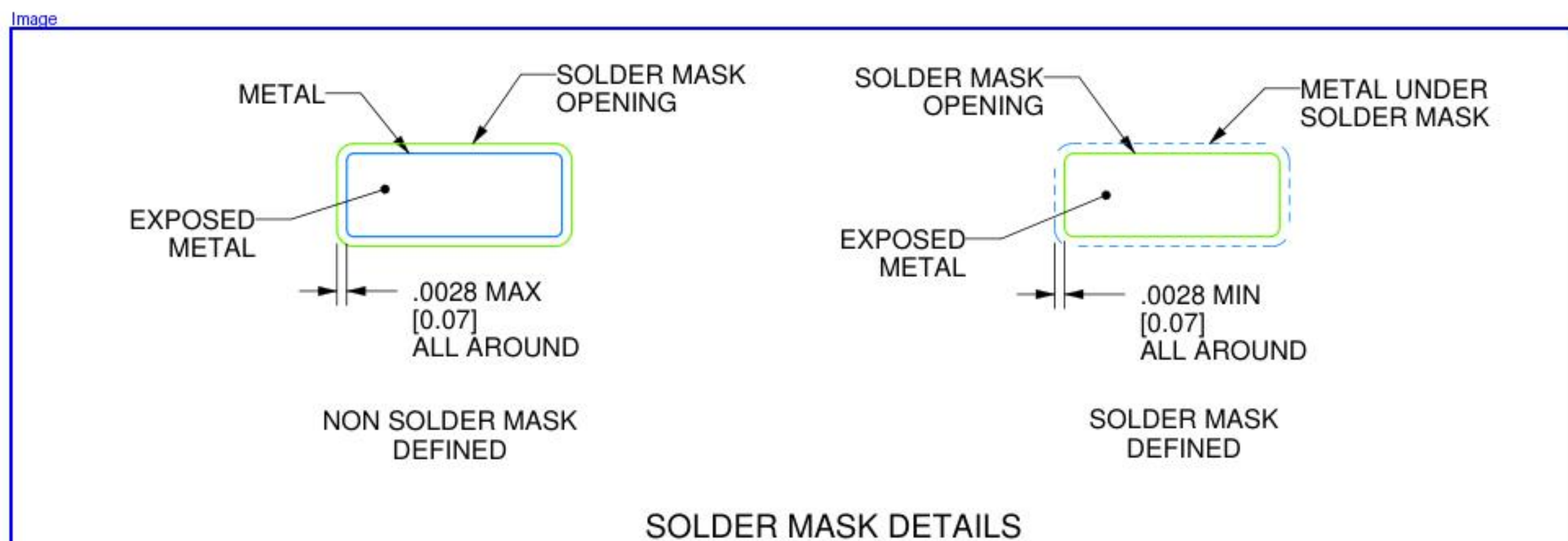
4. This dimension does not include interlead flash.

5. Reference JEDEC registration MS-012, variation AA.

Image



FigureCaption
LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:8X



4214825/C 02/2019

Narrative Text
NOTES: (continued)

List Item
6. Publication IPC-7351 may have alternate designs.

List Item
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

Title

EXAMPLE STENCIL DESIGN

Header

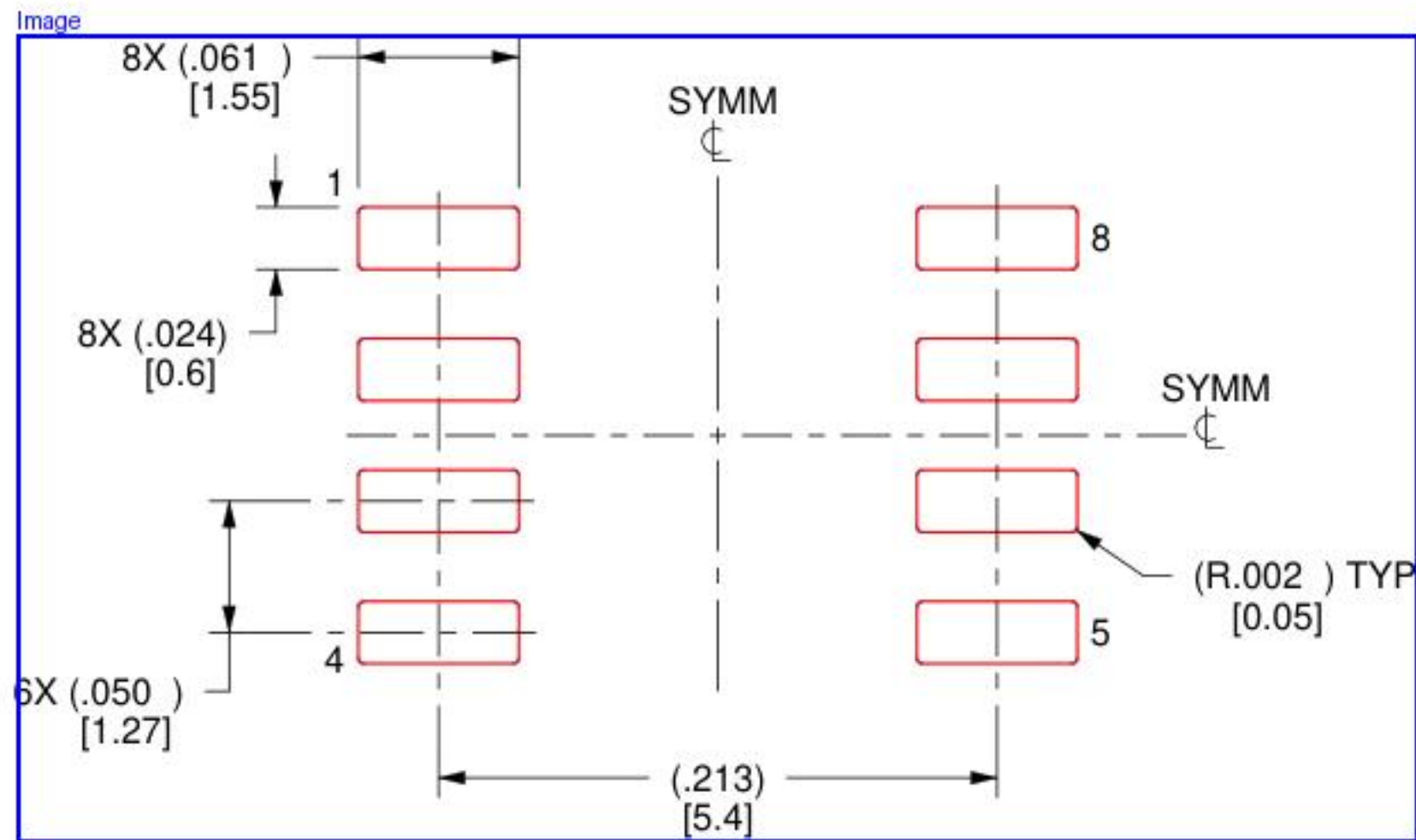
D0008A

Title

SOIC - 1.75 mm max height

Narrative Text

SMALL OUTLINE INTEGRATED CIRCUIT



FigureCaption

SOLDER PASTE EXAMPLE
 BASED ON .005 INCH [0.125 MM] THICK STENCIL
 SCALE:8X

Header

4214825/C 02/2019

Narrative Text

NOTES: (continued)

ListItem

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.

ListItem

9. Board assembly site may have different recommendations for stencil design.

Image

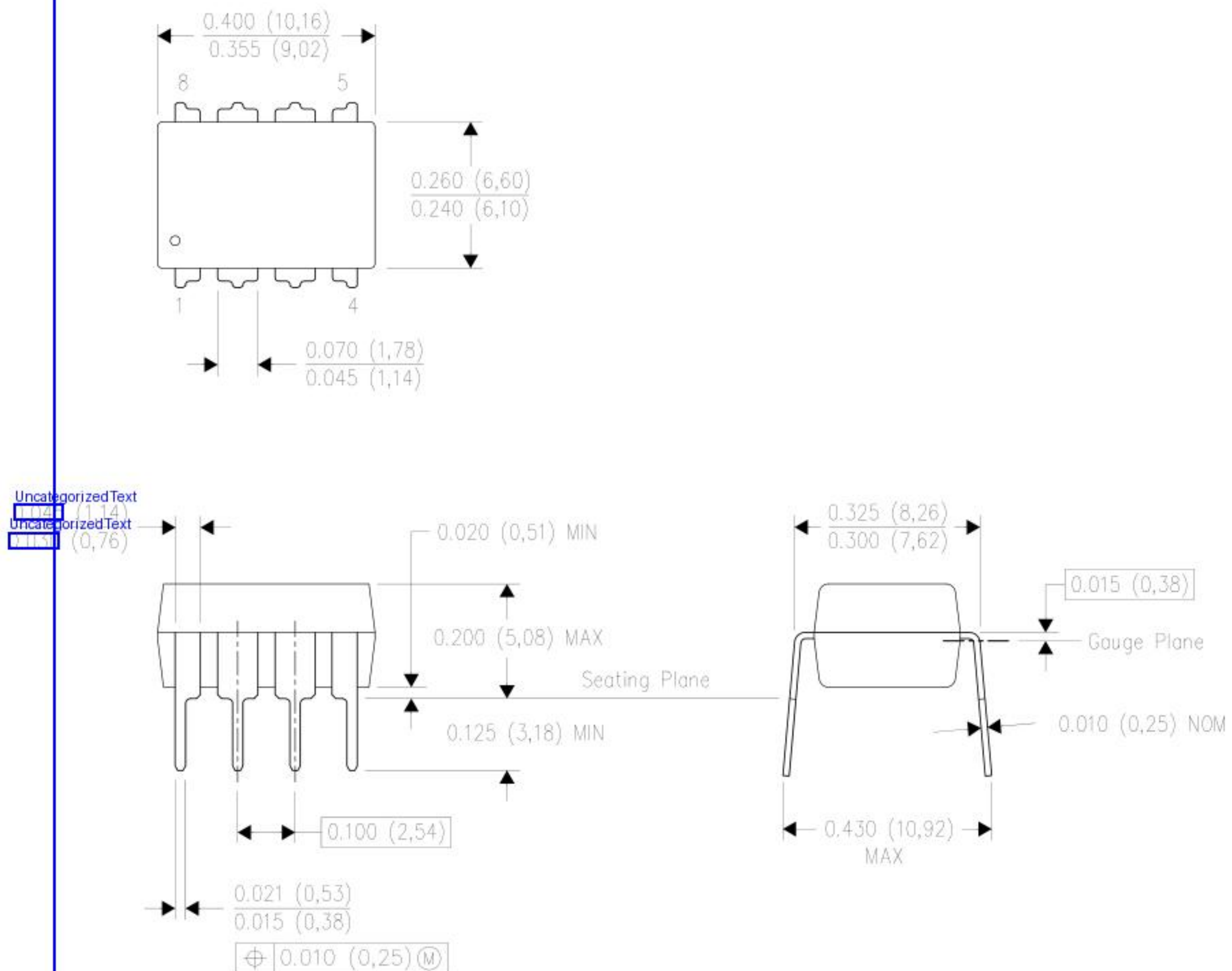
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P (R-PDIP-T8)

Narrative Text

PLASTIC DUAL-IN-LINE PACKAGE

Image



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4040082/E 04/2010

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NOTES

- All linear dimensions are in inches (millimeters).
- This drawing is subject to change without notice.
- Falls within JEDEC MS-001 variation BA.

Image

Title

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